CS 6362 Advanced Machine Learning, Fall 2026
Assignment 2: Let's read papers as a researcher and a reviewer

Hirak Sarkar

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1 Overview

Reading a paper submitted to a conference or journal needs a lot of attention and critical thinking. Since a submitted paper might not have all the details and not peer-reviewed yet, it might contain errors or mistakes. In this assignment, we will practice reading, **reproducing**, and critically reviewing a paper. Note the emphasis on **reproducing** the results, as this is a crucial skill for researchers. Given the enormous progress in LLMs, the implementation of a new paper has become easy. Though it might be tempting to rely completely on automated code generation, the generated results might not be generalizable. For example LLM might just generate code that looks like it works, but it might just reproduce the appearance of the plot (by hardcoding the values) rather than actually writing the correct code.

I have hand-picked 3 published such papers. You are required to complete all 3 papers, worth 50 points each for a total of 150 points. Submit one report per paper (three reports total), each following the rubric in Section 1.1. For each paper you will need to:

- Read and understand the model and the methodology, write a section summarizing your understanding.
- Follow my lecture style to describe complicated mathematical derivations and explanations, deriving them only when necessary, and provide intuition when appropriate.
- For each paper, we concentrate on one figure that you have to reproduce. You are **required to use an LLM** to reproduce it, through its **chat interface** (e.g., ChatGPT, Claude) rather than an API or IDE assistant, so that the full interaction is preserved. Use the LLM's built-in share-link feature to make the conversation viewable (ChatGPT: *Share* button on a chat; Claude: *Share chat*), and submit that link along with the name and version of the LLM (e.g., GPT-5, Claude Opus 4.5, Gemini 2.5 Pro). Also describe how your reproduction attempts might have failed. What mistakes or misunderstandings led to the failure? How have you corrected them? Use `pytorch` library for implementation.
- Critically review the paper, noting any strengths, weaknesses, and potential errors. This part does not need reproduction.

1.1 Grading rubrics for the report

As a result of your reading, reproducing, and reviewing the paper, your final report should include *following sections*:

1. Include a **problem statement** (Mathematically grounded with *definition of variables*). A summary of your understanding of the model and methodology. For explanation use necessary mathematical derivations when necessary and provide intuition when appropriate. **[10 pts]**
2. Reproduction of the chosen figure along with the original figure, including the shared conversation link, the name and version of the LLM, and a description of any failures and corrections. Please provide the `github` link to your reproduction code. Make it public on the day you submit the assignment (not before or after). The github repo should contain all the necessary code and instructions to reproduce the figure. A notebook (`.ipynb`) should be included to reproduce the figure. **Do not copy** an existing implementation yourself. During Phases 1–3, the LLM may not browse for or be given existing code. If it retrieves any anyway, you must disclose this in your report. **[25 pts]**
   - Phase 1: Give the paper pdf and prompt to the LLM.
   - Phase 2: Review the output from the LLM and identify any failures or inaccuracies.
   - Phase 3: Make a careful, good-faith reproduction attempt, correcting mistakes or misunderstandings as you find them. If your reproduction still does not match the original figure, diagnose *why* with evidence (e.g., ablations, intermediate checks, comparison to the paper's stated setup) rather than guessing. Save (commit) this version of your code before moving to Phase 4.
   - Phase 4: Only now may you consult an existing implementation, if the authors have released one on github. Compare it with your saved Phase 3 code and note any discrepancies. If you make further changes as a result, document exactly what you changed and why in your report.

   A technically sound, well-documented mismatch – where you clearly explain what you tried and why it likely diverges from the paper – can earn full credit. An unexplained failure to reproduce the figure cannot.
3. Write your own critical review of the paper first, noting strengths, weaknesses, and potential errors, then compare it with the paper's public OpenReview reviews. **[15 pts]**

2 Paper 1 (published) [50 pts]

Understanding deep learning requires rethinking generalization[1] by Zhang et al. 2017 [3] was accepted at ICLR 2017. It's a seminal paper that challenges the conventional understanding of generalization in deep learning, arguing that traditional theories may not fully explain why over-parameterized models can still generalize well. It's cited over 6000 times. The paper demonstrates that over-parameterized neural networks can fit training data even when labels or inputs are randomized, while still generalizing well when trained on real data.

Following the rubric outlined in Section 1.1, your goal is to reproduce **Figure 1 (panels a,b,c)** of the paper. Your reproduction should capture the experimental conditions and the qualitative behavior reported in the original figure, not merely produce visually similar plots.

Read the paper carefully before beginning the implementation. In addition to understanding the experiments in Figure 1, pay attention to how the authors use these experiments to build the paper's broader argument about optimization, memorization, and generalization. This exercise

[1] https://openreview.net/pdf?id=Sy8gdB9xx

is also intended to give you experience reading a seminal ML paper closely and translating its experimental description into an executable implementation.

Questions your report must answer:

- How are random labels generated across epochs, and what would go wrong otherwise?
- With corruption probability p, what fraction of labels actually differ from the truth, and what test error should you expect at $p = 1$?
- What optimizer, learning-rate schedule, preprocessing and regularization do the authors use?

Choices you must make and report:

- How you measure "convergence time" in Fig. 1b.
- Which corruption levels and architectures you run, if you reduce the grid.

3 Paper 2 (published) [50 pts]

On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima[2] by Keskar et al. [2] investigates the relationship between batch size and generalization gap in deep learning. It's also a seminal paper that has influenced subsequent research on the effects of batch size on model generalization. The paper demonstrates that larger batch sizes can lead to sharper minima in the loss landscape, which in turn can affect the generalization performance of the model.

Following the rubric outlined in Section 1.1, reproduce the F_1, C_1, and C_2 panels of **Figure 3**. These experiments compare the solutions obtained using small-batch and large-batch training by examining the cross entropy loss along a path (interpolation) between the two learned parameter vectors. The datasets used are MNIST and CIFAR-10, making it doable within a reasonable computational budget.

This paper is an important early attempt to connect the *dynamics of optimization* with the *generalization behavior* of deep networks. It provides a concrete empirical setting in which two optimization procedures can fit the training data well yet arrive at solutions with very different apparent loss geometry and test performance.

Questions your report must answer:

- Which networks and datasets correspond to panels F1, C1, C2?
- What range of α does Figure 3 use, and what quantities are plotted on each axis?
- What batch sizes and optimizer define the small-batch and large-batch runs?

Choices you must make and report:

- How batch-norm statistics are computed at interpolated weights.
- Whether the two runs share an initialization.

[2] https://openreview.net/pdf?id=H1oyRlYgg

4 Paper 3: β-VAE (published) [50 pts]

β-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework [3] by Higgins et al. [1] was published at ICLR 2017. The paper asks whether a generative model can learn *disentangled* representations without supervision. A representation is disentangled when different latent variables capture distinct underlying factors of variation in the data. For example, one latent variable might represent horizontal position, another scale, and another rotation.

Following the rubric outlined in Section 1.1, reproduce **Figure 7A and 7B**. These panels compare a β-VAE ($\beta = 4$) against a standard VAE ($\beta = 1$) on the 2D Shapes (dSprites) dataset. The dataset and its known generative factors are publicly available from the official dSprites repository. Your reproduction should capture both how individual latent variables respond to the known generative factors and what happens to the decoded image when individual latent variables are traversed while the others are held fixed.

The central claim being tested is that the stronger KL constraint used by the β-VAE ($\beta = 4$), compared with the standard VAE ($\beta = 1$), produces a more disentangled latent representation. Your analysis should explain what evidence Figure 7A,B provides for this claim and whether your reproduction exhibits the same qualitative difference between the two models. Do you think increasing β further would affect the disentanglement of the latent representation, and what would happen to reconstruction quality?

Questions your report must answer:

- What happens to L_β for $\beta > 1$? For $\beta < 1$?
- How are the latents in Figure 7 ordered, and how is each traversal generated?

Choices you must make and report:

- How many seeds per β. Say what one seed can and cannot show.

References

[1] Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, and Alexander Lerchner. beta-vae: Learning basic visual concepts with a constrained variational framework. In *International conference on learning representations*, 2017.

[2] Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations*, 2017.

[3] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. In *International Conference on Learning Representations*, 2017.

[3] https://openreview.net/pdf?id=Sy2fzU9gl